1. Introduction

This report presents a comprehensive **Power BI dashboard** developed for **ShopNest Store** to analyze key business performance metrics using retail transaction data.
The dashboard integrates multiple datasets related to orders, products, customers, payments, and reviews to provide actionable insights for business decision-making.

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2. Data Overview

The analysis uses the following datasets:

- Orders
- Order Items
- Products
- Customers
- Payments
- Reviews
- Sellers

Relationships were created using common keys such as **order_id, product_id, and customer_id**, enabling integrated analysis across sales, delivery performance, customer behavior, and revenue trends.

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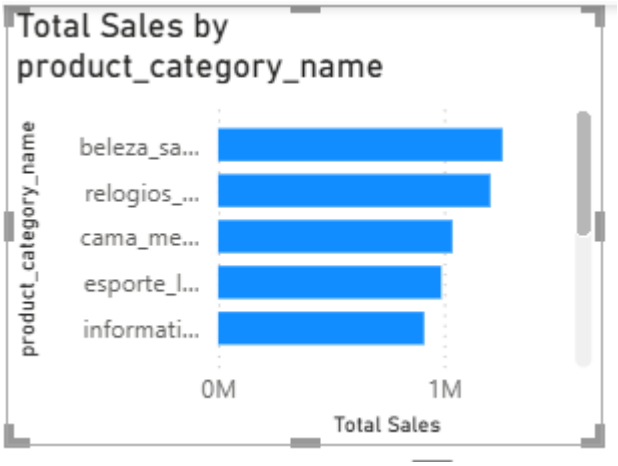
3. Question 1: Top Categories by Total Price

Question Statement

Identify and visually represent the **top 10 product categories by total sales**.

Visualization



Explanation

This bar chart displays the **top 10 product categories** based on total sales value.
Categories such as electronics, home appliances, and personal care contribute significantly to overall revenue.
This insight helps management focus on **high-performing categories** for inventory optimization and targeted promotions.

Key Insights & Findings

Key Insights & Findings

- **Information Technology Accessories** and **Health & Beauty** categories contribute the **highest total sales**, clearly outperforming other categories.
- Sales are **heavily concentrated in a few categories**, indicating that revenue is driven by limited high-performing segments.
- Lower-ranked categories contribute significantly less, suggesting scope for **category-level promotions or rationalization**.

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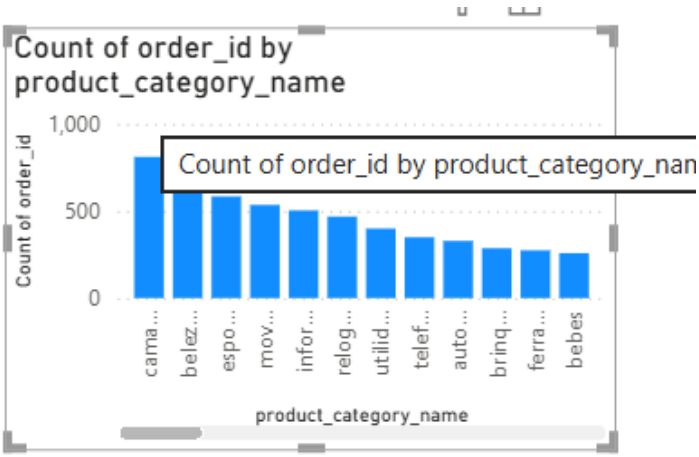
4. Question 2: Delayed Orders Analysis (Category-wise)

Question Statement

Determine the **number of delayed orders in each product category**.

Visualization



Explanation

This visualization shows the **count of delayed orders** across different product categories.
Categories with higher delayed order counts indicate potential **logistics or supply chain challenges**.
The analysis helps identify areas where delivery processes can be improved to enhance customer satisfaction.

Key Insights & Findings

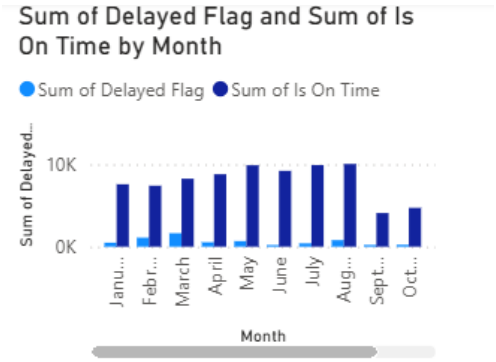
- Categories with **high order volumes** also tend to show **higher delayed order counts**, indicating operational pressure.
- Product categories such as **computers, electronics, and home-related items** experience more delays, possibly due to handling or logistics complexity.
- Improving delivery processes in these categories can significantly **reduce overall delivery delays**.

5. Question 3: Monthly Comparison of Delayed and On-Time Orders

Question Statement

Compare **delayed and on-time orders on a monthly basis**.

Visualization



Explanation

This column chart compares **monthly delayed and on-time deliveries**, highlighting variations in delivery performance over time.
Months with higher delayed orders indicate operational pressure periods, enabling the business to proactively improve logistics planning.

Key Insights & Findings

- On-time deliveries consistently remain **higher than delayed orders**, indicating overall healthy logistics performance.
- A noticeable **increase in delayed orders during mid-year months (May–July)** suggests seasonal or demand-related stress on delivery systems.
- Delivery performance improves in later months, indicating **better capacity planning or reduced order load**.

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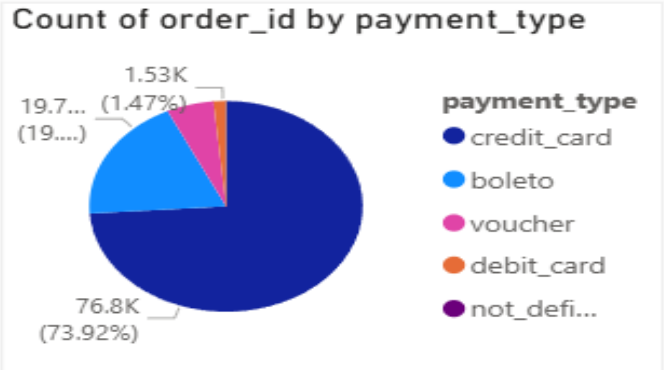
6. Question 4: Payment Method Analysis

Question Statement

Analyze the **most frequently used payment methods** by customers.

Visualization



Explanation

The pie chart represents the **distribution of payment methods** used by customers.
Credit cards emerge as the most preferred payment option, followed by digital and alternative methods.
This insight supports decisions related to payment partnerships and customer checkout experience optimization.

Key Insights & Findings

- **Credit cards dominate customer payments**, accounting for nearly **three-fourths of all transactions**.
- Boleto is the second most-used payment method, while debit cards and vouchers have minimal usage.
- This highlights strong customer trust in **digital and card-based payment systems**, suggesting low dependence on cash alternatives.

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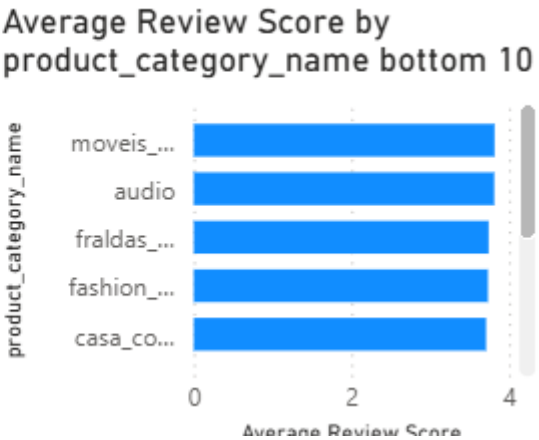
7. Question 5: Product Rating Analysis

Question Statement

Identify the **top 10 highest-rated** and **bottom 10 lowest-rated** products/categories.

Visualization



Explanation

These visuals highlight product categories based on **average customer review scores**.
High-rated categories indicate strong customer satisfaction, while low-rated categories reveal quality gaps or

unmet customer expectations.
This analysis helps prioritize **product improvements and quality control initiatives**.

Key Insights & Findings

Top Rated Categories

- Categories such as **CDs/DVDs, fashion items, and books** receive consistently **high average review scores**.
- These categories demonstrate strong customer satisfaction and product quality.

Bottom Rated Categories

- Categories like **movies, audio products, and fashion accessories** show relatively **lower average ratings**.
- Lower ratings indicate possible **quality gaps, unmet expectations, or post-purchase issues**, requiring focused improvement.

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8. Question 6: State-wise Sales Analysis

Question Statement

Analyze **sales performance across different states**.

Visualization

Explanation

The map visual represents **state-wise sales distribution**, where bubble size indicates total sales.
States with larger bubbles reflect stronger market presence, while smaller bubbles identify regions with potential growth opportunities.
This supports regional sales strategy and market expansion planning.

Key Insights & Findings

- Sales are **geographically concentrated in specific states**, indicating strong regional markets.

- A few states show **significantly higher sales volumes**, while several regions have low penetration.
- The distribution suggests opportunities for **regional marketing expansion and targeted growth strategies**.

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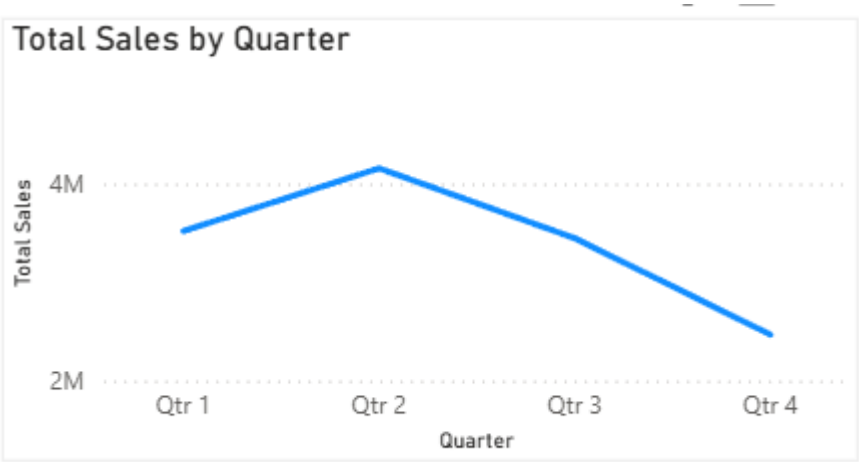
9. Question 7: Seasonal Sales Patterns (Quarterly)

Question Statement

Identify **seasonal trends in sales on a quarterly basis**.

Visualization



Explanation

This line chart shows **quarter-wise sales trends**, revealing seasonal patterns.
Sales peak in specific quarters, indicating high-demand periods, while lower quarters suggest off-season phases.
These insights assist in **inventory planning, staffing, and promotional timing**.

Key Insights & Findings

- **Quarter 2 records the highest sales**, indicating a peak seasonal demand period.
- Sales decline gradually from **Quarter 3 to Quarter 4**, highlighting an off-peak phase.
- Understanding this seasonal pattern can help optimize **inventory stocking and promotional timing**.

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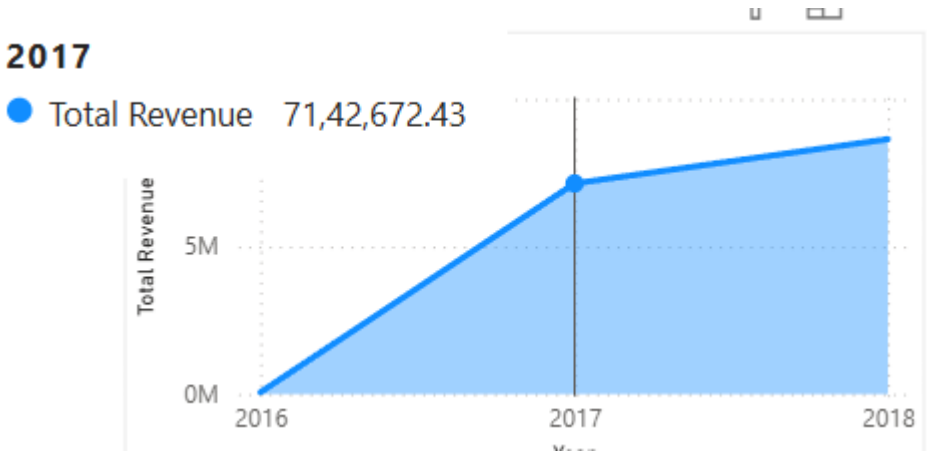
10. Question 8: Revenue Analysis (Yearly)

Question Statement

Analyze **total revenue growth over the years**.

Visualization



Explanation

The area chart illustrates **year-wise revenue growth** for ShopNest Store.
A consistent upward trend indicates improving business performance and successful scaling strategies.
This view supports long-term strategic planning and financial forecasting.

Key Insights & Findings

- Total revenue shows a **strong upward trend from 2016 to 2018**, reflecting consistent business growth.
- The sharp increase between **2016 and 2017** indicates rapid expansion or improved sales performance.
- Continued growth in 2018 confirms **business scalability and increasing market presence**.

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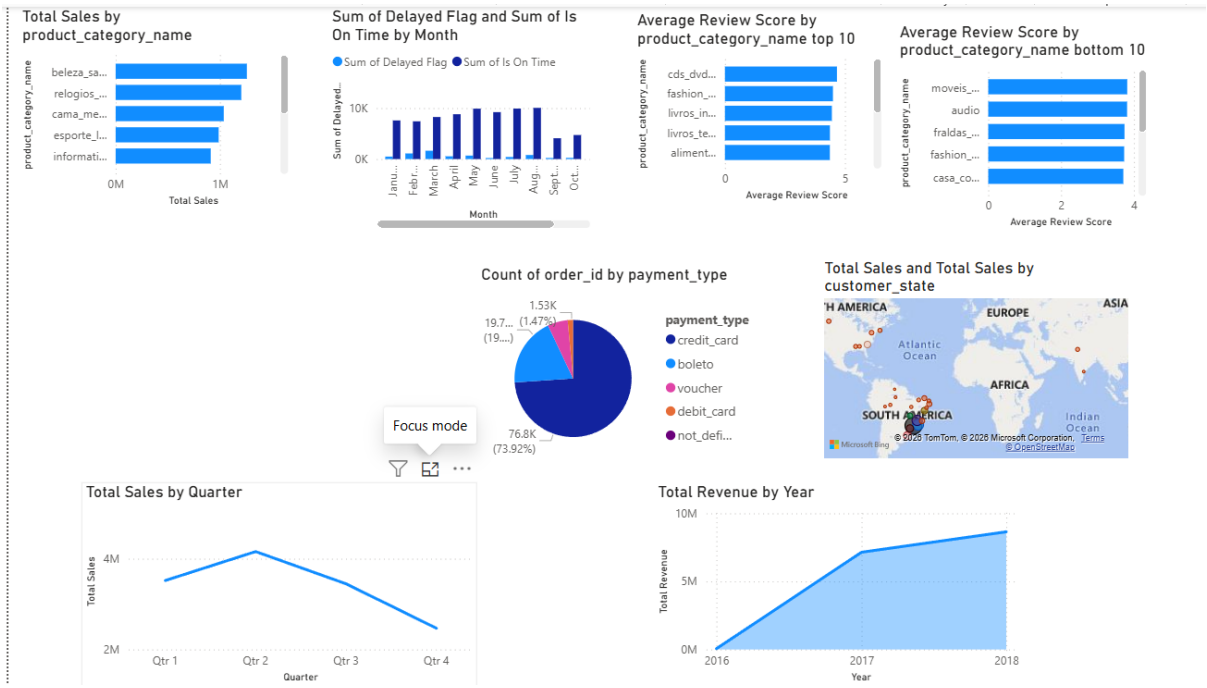
11. Conclusion

The ShopNest Store Power BI dashboard delivers a **holistic view of business performance**, covering sales, revenue, delivery efficiency, customer preferences, and regional trends.
The insights derived from this dashboard enable **data-driven decision-making**, operational improvements, and strategic growth planning.


Total Sales by product_category_name
product_category_name
beleza_sa...
relogios_...
cama_me...
esporte_l...
informati...
0M
1M
Total Sales
Sum of Delayed Flag and Sum of Is On Time by Month
Sum of Delayed Flag
Sum of Is On Time
Sum of Delayed...
10K
0K
Janu...
Feb...
March
April
May
June
July
Aug...
Sep...
Oct...
Month
Average Review Score by product_category_name top 10
product_category_name
cds_dvd...
fashion_...
livros_in...
livros_te...
aliment...
0
5
Average Review Score
Average Review Score by product_category_name bottom 10
product_category_name
moveis_...
audio
fraldas_...
fashion_...
casa_co...
0
2
4
Average Review Score
Count of order_id by payment_type
1.53K
(1.47%)
19.7...
(19....)
76.8K
(73.92%)
payment_type
credit_card
boleto
voucher
debit_card
not_defi...
Focus mode
Total Sales and Total Sales by customer_state
EUROPE
ASIA
Atlantic Ocean
AFRICA
Indian Ocean
© 2025 TomTom, © 2026 Microsoft Corporation, Terms
© OpenStreetMap
Total Sales by Quarter
Total Sales
4M
2M
Qtr 1
Qtr 2
Qtr 3
Qtr 4
Quarter
Total Revenue by Year
Total Revenue
10M
5M
0M
2016
2017
2018
Year